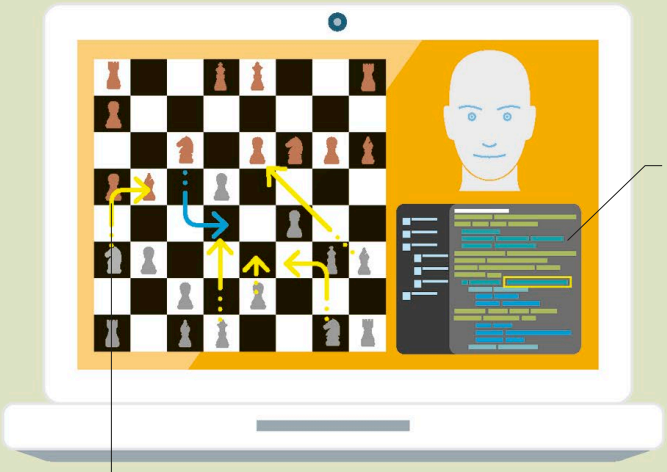
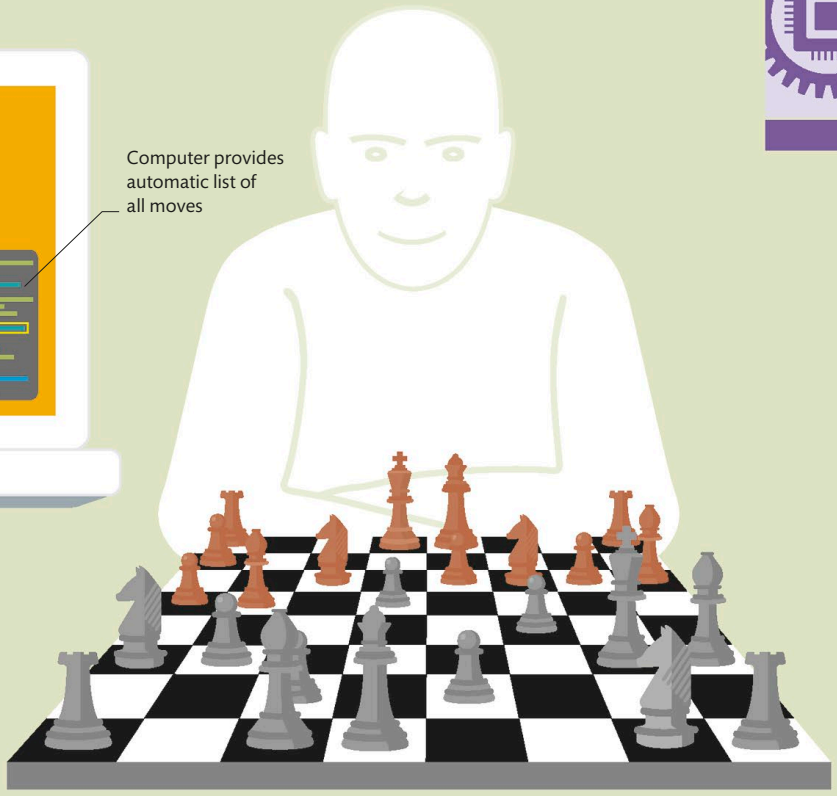


COMPUTER CHESS PLAYER

Human versus computer
The human brain can look only a few moves ahead, while emotions and "gut feelings" may help, or sometimes hinder, the player. A computer looks at all the possible moves then picks the one that seems most promising. It can look many moves ahead for each scenario.



HUMAN CHESS PLAYER

Playing games

Computers with AI can play games that require intelligence for humans to play, including complicated games such as chess. Powerful chess computers have even beaten the world's best human chess players. However, a game-playing computer can work only within the rules of the game; if anything happens that is outside the rules, the computer is unable to respond. Most game-playing computers follow programs that help them make the best move by analyzing all the possible moves and likely outcomes. In combination with machine learning (see opposite), AI systems can improve their skills at games.



IN 1997, THE COMPUTER **DEEP BLUE** FIRST BEAT THE CHESS WORLD CHAMPION **GARRY KASPAROV**

APPLICATIONS OF AI

Suggestions for music based on recent listening
Machine learning finds songs that people with similar music tastes have chosen.

Best route for package deliveries
In conjunction with digitized maps and traffic patterns, AI systems can save time and increase efficiency.

Helping doctors diagnose illnesses
Fed a patient's symptoms, an AI system can search medical databases to suggest possible causes.

Self-driving cars
Computers fed with images from onboard cameras, radar, and digital maps can drive cars safely.

Filtering out spam emails
Instead of simply blocking certain senders' addresses, this system recognizes patterns and adapts to new trends.

Image recognition
Artificial neural networks improve recognition of objects in digital images, even if the image is unclear.